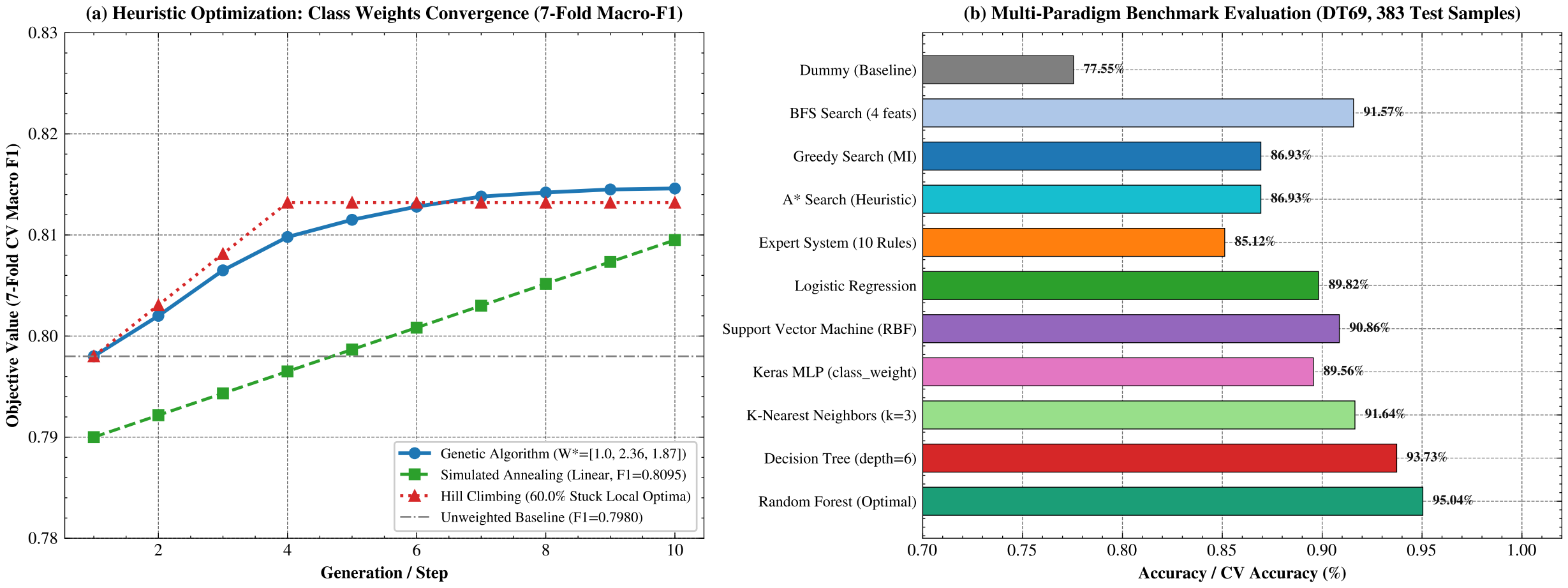
(a) Heuristic Optimization: Class Weights Convergence (7-Fold Macro-F1)
Objective Value (7-Fold CV Macro F1)
Generation / Step
0.83
0.82
0.81
0.80
0.79
0.78
2
4
6
8
10
Genetic Algorithm (W*=[1.0, 2.36, 1.87])
Simulated Annealing (Linear, F1=0.8095)
Hill Climbing (60.0% Stuck Local Optima)
Unweighted Baseline (F1=0.7980)
(b) Multi-Paradigm Benchmark Evaluation (DT69, 383 Test Samples)
Dummy (Baseline)
77.55%
BFS Search (4 feats)
91.57%
Greedy Search (MI)
86.93%
A* Search (Heuristic)
86.93%
Expert System (10 Rules)
85.12%
Logistic Regression
89.82%
Support Vector Machine (RBF)
90.86%
Keras MLP (class_weight)
89.56%
K-Nearest Neighbors (k=3)
91.64%
Decision Tree (depth=6)
93.73%
Random Forest (Optimal)
95.04%
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Accuracy / CV Accuracy (%)